Empirically grounded model of observable management-control practices (WBES, G7, 2024-2025)

Exploratory, cross-sectional, firms with the management module (n = 6,316; mostly 20+ employees). Associations only - no causal claims.

LAYER 1 - Observable practices (WBES r1-r11)

MONITORING
r2 performance indicators tracked (y/n)
r3 number of indicators (given r2)
r1 reaction to production problems

V=.41 (r2-r4)

TARGETS
r4 targets exist (y/n)
r5 time frame of targets (given r4)
r6, r7 other target features
(wording NOT verified)

V=.22-.27

INCENTIVES
r8 bonuses tied to targets (y/n)
r9 basis of bonuses (given r8)

V=.15-.26

PEOPLE-RELATED ITEMS
r10, r11 (wording NOT verified;
incl. 'does-not-apply' codes;
comparability across countries doubtful)

LAYER 2 - Empirical structure (statistically identified)

ONE DOMINANT CONTINUOUS DIMENSION
'breadth of formalised practice'

Parallel analysis retains 1 factor (6 items:
eigenvalue 2.54 vs 1.09)
Marginal reliability of the trait = 0.60
Tucker congruence across countries .93-.99

DISCRETE PROFILES: WEAK
LCA entropy .53-.60 (spec A, K=3-4); min. AvePP .64-.70
BIC flat for K = 3-5; class order = gradient (K=3)
K=4: small class (5-7%) with targets but fewer
bonuses - unstable across subsamples

COMBINATIONS
All-three gates present: 43% (1.35x expected)
None present: 10% (3.7x expected)
Most partial combinations rarer than expected
Country composition differs (e.g. none: ITA 27%)
Some items non-invariant across countries

BIC across families: hybrid (3 classes x 1 trait) < 2-D continuous < 1-D continuous < LCA (spec A) | 2-D continuous best in B, C, D -> 'hybrid'/multi-dimensional, NOT pure typology

associations with the continuous score

LAYER 3 - Theoretical interpretation (dashed = not tested here)

MALMI & BROWN (2008)
Observed items mainly map to cybernetic
(monitoring, targets) and reward controls;
cultural, planning, administrative largely
unobserved. Package coexistence: supported
as co-occurrence only

SIMONS (1995)
Monitoring/targets are COMPATIBLE with
diagnostic control. Interactive use, belief
and boundary systems are NOT observable
in WBES -> classes are not labelled
'diagnostic' or 'interactive'

BLOOM & VAN REENEN (2007)
An aggregate score captures most structure
in the data (gradient); configurations add
little once the score is controlled in the
outcome models examined

LAYER 4 - Organisational associations (cross-sectional)

FIRM CHARACTERISTICS
Larger firms: higher mean core score
(1.86 -> 2.51, <50 to 250+ staff)
Country gaps: ITA lowest, GBR highest

PERFORMANCE (associational)
log sales/employee: +0.13 per SD of score (within country)
3y sales growth: +0.003 per SD; employment growth: +0.002
Classes add nothing once the continuous score is controlled
(pseudo-class F tests, all outcomes)

NONLINEARITY / INTERACTION
Productivity: mild non-linearity (Holm p<.05)
No robust interactions among monitoring,
targets, bonuses (Holm)

Solid: estimated structure/association. Dashed: theoretical reading, not tested. No arrow implies causation.